

## LA CEIBA GOLOSON, Honduras

WMO: 787050

Lat: 15.742N

Lon: 86.853W

Elev: 49

StdP: 14.67

Time Zone: -6.00 (W06)

Period: 95-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 63.4       | 64.7 | 60.5                          | 78.7 | 66.6 | 62.3 | 84.0 | 67.5 | 18.5                  | 75.9 | 16.4 | 76.6 | 2.3                   | 190  | 0.264 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 6             | 14.2                   | 91.0            | 80.1 | 89.9 | 79.7 | 89.4 | 79.6 | 81.8                | 88.3 | 81.0 | 87.8 | 80.5 | 87.4 | 9.0                  | 30   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 80.0                            | 155.8 | 86.6 | 79.1 | 151.2 | 86.0 | 78.7 | 149.4 | 85.7 | 45.6          | 89.0 | 44.8 | 88.3 | 44.3 | 87.2 | 84.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|--|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |  |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |  |
| 17.8              | 15.2 | 12.9 | DB                         | 59.1 | 93.7               | 2.9 | 2.8                                                | 57.0 | 95.7       | 55.2 | 97.3       | 53.6 | 98.8       | 51.5 | 100.8 |  |
|                   |      |      | WB                         | 57.7 | 82.7               | 2.3 | 0.9                                                | 56.0 | 83.3       | 54.6 | 83.8       | 53.3 | 84.3       | 51.6 | 85.0  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 78.0   | 73.5  | 75.1  | 75.8  | 78.6  | 80.6  | 81.3  | 80.8  | 80.8  | 81.1  | 78.9  | 75.4  | 74.3  |
|                                                                        | DBStd       | 3.64   | 2.83  | 2.28  | 2.76  | 2.78  | 2.01  | 1.54  | 1.24  | 1.15  | 1.28  | 2.47  | 2.83  | 2.74  |
|                                                                        | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | CDD50       | 10236  | 729   | 704   | 801   | 859   | 950   | 939   | 955   | 956   | 933   | 895   | 761   | 754   |
|                                                                        | CDD65       | 4762   | 265   | 284   | 336   | 409   | 485   | 489   | 490   | 491   | 483   | 430   | 311   | 289   |
|                                                                        | CDH74       | 42047  | 1423  | 2017  | 2784  | 4005  | 5073  | 5107  | 5020  | 5002  | 4842  | 3457  | 1750  | 1567  |
|                                                                        | CDH80       | 15070  | 255   | 483   | 823   | 1468  | 2035  | 2028  | 1982  | 2073  | 1991  | 1215  | 407   | 310   |
| (14) Wind                                                              | WSAvg       | 4.4    | 4.1   | 4.2   | 4.7   | 4.4   | 4.6   | 4.9   | 5.8   | 4.6   | 3.8   | 3.7   | 3.8   | 3.6   |
| (15) Precipitation                                                     | PrecAvg     | 83.6   | 10.1  | 6.5   | 5.4   | 1.8   | 3.1   | 5.2   | 4.0   | 5.4   | 7.7   | 11.1  | 12.8  | 8.9   |
|                                                                        | PrecMax     | 133.0  | 25.6  | 18.5  | 16.9  | 5.3   | 7.4   | 10.2  | 9.0   | 10.7  | 18.0  | 30.4  | 33.2  | 28.7  |
|                                                                        | PrecMin     | 44.4   | 1.2   | 0.3   | 0.0   | 0.0   | 0.1   | 0.9   | 0.4   | 0.7   | 1.3   | 0.0   | 0.6   | 0.0   |
|                                                                        | PrecStd     | 23.9   | 7.3   | 4.4   | 4.5   | 1.7   | 2.0   | 2.5   | 2.2   | 2.7   | 4.1   | 7.6   | 8.1   | 6.5   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 85.7  | 86.4  | 89.5  | 91.3  | 91.6  | 91.6  | 90.6  | 91.3  | 91.6  | 90.0  | 88.0  | 86.4  |
|                                                                        |             | MCWB   | 76.2  | 76.9  | 77.0  | 80.4  | 79.6  | 80.6  | 79.1  | 81.0  | 80.5  | 80.3  | 78.6  | 77.3  |
|                                                                        | 2%          | DB     | 84.2  | 85.5  | 86.9  | 89.2  | 89.8  | 90.0  | 89.7  | 89.8  | 89.9  | 89.0  | 86.1  | 84.6  |
|                                                                        |             | MCWB   | 75.7  | 76.4  | 77.3  | 79.7  | 79.6  | 79.9  | 79.0  | 79.9  | 80.0  | 79.9  | 78.1  | 76.2  |
|                                                                        | 5%          | DB     | 82.7  | 84.3  | 85.9  | 87.7  | 89.1  | 89.4  | 88.6  | 89.3  | 89.4  | 87.8  | 84.4  | 83.1  |
|                                                                        |             | MCWB   | 74.9  | 75.9  | 76.9  | 79.0  | 79.4  | 79.7  | 79.0  | 79.7  | 79.9  | 79.3  | 76.9  | 75.5  |
|                                                                        | 10%         | DB     | 81.7  | 83.1  | 84.4  | 86.3  | 87.9  | 88.0  | 87.8  | 88.0  | 88.1  | 86.4  | 82.7  | 82.1  |
|                                                                        |             | MCWB   | 74.4  | 75.2  | 76.2  | 78.1  | 79.1  | 79.3  | 78.9  | 79.3  | 79.5  | 78.4  | 75.9  | 75.0  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 78.3  | 79.1  | 80.0  | 82.5  | 81.9  | 82.3  | 81.1  | 82.1  | 82.5  | 82.3  | 80.0  | 78.7  |
|                                                                        |             | MCDB   | 83.1  | 84.4  | 85.9  | 87.9  | 88.5  | 89.7  | 87.6  | 88.8  | 88.6  | 88.4  | 84.5  | 84.6  |
|                                                                        | 2%          | WB     | 76.7  | 77.9  | 78.8  | 80.8  | 80.8  | 81.2  | 80.3  | 81.1  | 81.4  | 80.7  | 78.9  | 77.3  |
|                                                                        |             | MCDB   | 82.2  | 83.7  | 85.0  | 87.3  | 87.8  | 88.2  | 87.1  | 88.0  | 88.1  | 87.3  | 84.6  | 83.0  |
|                                                                        | 5%          | WB     | 75.7  | 76.6  | 77.9  | 79.7  | 80.1  | 80.4  | 79.8  | 80.3  | 80.6  | 79.7  | 77.8  | 76.4  |
|                                                                        |             | MCDB   | 81.5  | 82.7  | 84.2  | 86.3  | 87.1  | 87.2  | 86.7  | 87.3  | 87.7  | 86.3  | 83.4  | 81.9  |
|                                                                        | 10%         | WB     | 74.7  | 75.8  | 76.8  | 78.8  | 79.4  | 79.8  | 79.3  | 79.8  | 79.9  | 78.9  | 76.6  | 75.4  |
|                                                                        |             | MCDB   | 80.3  | 81.8  | 83.3  | 85.3  | 86.4  | 86.5  | 86.1  | 86.7  | 86.9  | 85.3  | 81.9  | 80.7  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 13.2  | 15.3  | 16.2  | 15.7  | 14.9  | 14.2  | 15.1  | 15.4  | 15.1  | 13.6  | 12.7  | 13.1  |
|                                                                        |             | MCDBR  | 16.2  | 17.0  | 17.9  | 16.2  | 15.7  | 15.4  | 16.0  | 16.6  | 16.1  | 15.8  | 15.7  | 16.0  |
|                                                                        |             | MCWBR  | 9.5   | 9.8   | 9.9   | 8.7   | 7.8   | 7.5   | 7.9   | 8.5   | 8.2   | 8.4   | 9.0   | 9.0   |
|                                                                        | 5% WB       | MCDBR  | 15.6  | 16.5  | 17.1  | 15.4  | 14.8  | 14.5  | 15.3  | 15.7  | 15.3  | 15.0  | 14.7  | 15.2  |
|                                                                        |             | MCWBR  | 9.7   | 10.0  | 10.2  | 8.9   | 7.9   | 7.6   | 8.0   | 8.5   | 8.2   | 8.4   | 9.0   | 9.0   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.378 | 0.386 | 0.402 | 0.482 | 0.511 | 0.464 | 0.472 | 0.449 | 0.428 | 0.409 | 0.397 | 0.401 |
|                                                                        | taud        |        | 2.549 | 2.519 | 2.464 | 2.209 | 2.154 | 2.340 | 2.296 | 2.380 | 2.433 | 2.502 | 2.517 | 2.497 |
|                                                                        | Ebn at Noon |        | 290   | 292   | 289   | 265   | 254   | 263   | 262   | 271   | 278   | 281   | 282   | 279   |
|                                                                        | Edh at Noon |        | 31    | 34    | 36    | 47    | 49    | 40    | 42    | 39    | 37    | 34    | 32    | 32    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1234  | 1555  | 1803  | 1885  | 1775  | 1735  | 1777  | 1827  | 1753  | 1371  | 1178  | 1165  |
|                                                                        | RadStd      |        | 149   | 140   | 142   | 82    | 105   | 103   | 78    | 89    | 82    | 171   | 124   | 119   |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (4 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page